

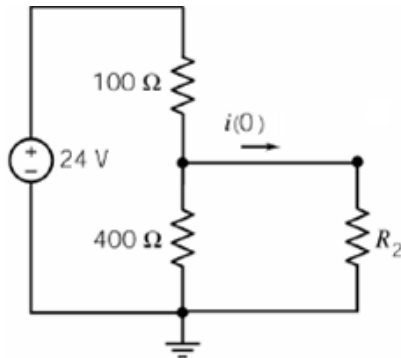
1. Do the problem in 3 steps.

Do you remember this equation?

$$x(t) = x_f + (x_0 - x_f)e^{-\frac{t}{\tau}}$$

$$i(t) = i(\infty) + (i(0) - i(\infty))e^{-\frac{t}{\tau}}$$

Step 1. Draw the circuit before $t = 0$ to determine $i(0)$



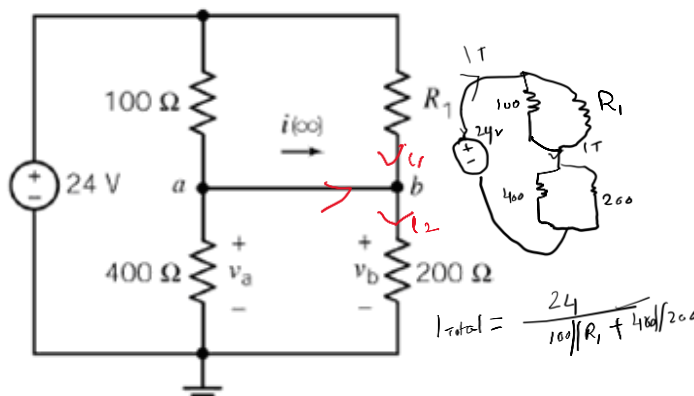
$$i(0) - i(\infty) = 53.6 \text{ mA} \quad i(0) = 68.6 \text{ mA}$$

$$i(t) = 15 + 53.6e^{-548t} \text{ mA}$$

$$0.0686 = i(0) = \left(\frac{400}{400 + R_2} \right) \left(\frac{24}{100 + (400 \parallel R_2)} \right)$$

$$R_2 = 199.988 \approx 200 \Omega$$

Step 2. Draw the circuit at $t = \infty$ to determine $i(\infty)$

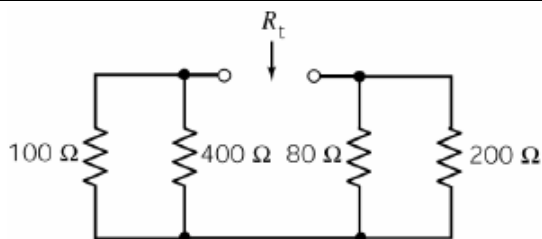


$$i(t) = 15 + 53.6e^{-548t} \text{ mA}$$

$$i(\infty) = \lim_{t \rightarrow \infty} i(t) = 0.015$$

$$R_1 = 80 \Omega$$

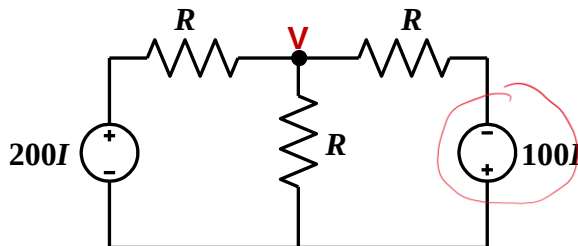
Step 3. Draw the circuit to determine Thevenin's resistance



$$R_t = (400 \parallel 100) + (80 \parallel 200) = 80 + 57 = 137 \Omega$$

$$\frac{L}{R_t} = \frac{L}{137} = \frac{1}{548} \Rightarrow L = 0.25 \text{ H}$$

2. Check the polarity of the mmf 100 I. If this is not correct and other calculations are correct you will be awarded only 4 marks.



$$\text{mmf} = NI$$

$$\frac{\text{mmf}}{\text{reluctance}} = \text{flux}$$

$$A = (20+1)(10+1) = 231 \text{ mm}^2$$

$$R = \frac{l}{\mu A} = \frac{0.001}{4\pi \times 10^{-7} \times 0.000231} = 3444912.2 \text{ A/wb} \quad \boxed{2}$$

mmf — V
flux — I
Rel. = Resistance

From the figure node equation is $\frac{V-200I}{R} + \frac{V+100I}{R} + \frac{V}{R} = 0$, $3V = 100I$, $V = \frac{100}{3}I$

Current in the middle branch is $\frac{V}{R} = \frac{100}{3R}I = 8 \times 10^{-5}$

So the current I is $I = \frac{3R \times 8 \times 10^{-5}}{100} = \frac{3 \times 3444912.2 \times 8 \times 10^{-5}}{100} = 8.2678 \text{ A}$

(3)

Natural frequency is $\omega_n = \sqrt{2}$ rad/sec. At ω_n , the phase is 90 degree. Since 3 rad/sec > ω_n , the phase at 3 rad/sec will be between 90 and 180 degree.

$$G(s) = \frac{Z_2(s)}{Z_1(s) + Z_2(s)} = \frac{2/s}{2 + s + 2/s} = \frac{2}{s^2 + 2s + 2} \quad 2 \text{ marks}$$

$$\omega = 2\pi f = 2\pi \times \frac{1}{T} = 2\pi \times \frac{1}{1.047 \times 2} = 3 \text{ rad/sec} \quad 3 \text{ marks}$$

$$V_i = 2 \sin 3t \quad G(j\omega) = \frac{2}{(2-\omega^2) + j2\omega} = \frac{2}{-7 + j6} \quad M \angle \phi$$

$$M = \frac{2}{\sqrt{49+36}} = 0.2169 \quad \phi = -139.4^\circ$$

$$V_o(t) = 2 \times 0.2169 \sin(3t - 139.4^\circ) \quad 5 \text{ marks}$$

(4)

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} = \frac{25}{s^2 + 5s + 25}$$

$$\omega_n = 5$$

$$\zeta = 0.5$$

$$\text{Time constant} = \frac{1}{\zeta\omega_n} = 0.4$$

$$\text{Roots} = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2}$$

The complex roots are in the LHS plane. The response stable because the roots are in the left half of s-plane. Oscillatory because the poles are complex. The amplitude of the oscillations will continue to decrease after some time (after 6 time constants approx), the response will reach steady state.

$$V_o(s) = \frac{1}{s} \times \frac{25}{s^2 + 2s + 25}$$

Initial and final value theorem

steady state value $\lim_{s \rightarrow 0} s \cdot V(s)$

$$s \rightarrow 0$$

$$= \frac{s}{s} \cdot \frac{25}{s^2 + 2s + 25} = 1$$

5. If you know that modes (eigenvalues appear in the output) appear in the output, you would have saved time. The second eigenvalue is not appearing because of the unique value of C and $X(0)$.

One of the eigenvalue is -1

$$y = \underline{C} e^{-t}$$

$$|sI - A| = s^2 + 2s + 2 = 0$$

Let the second eigenvalue is $-\lambda_2$

$$(s+1)(s+\lambda_2) = 0 \quad s^2 + (1+\lambda_2)s + \lambda_2 = 0$$

$\lambda_2 = 2$ eigenvalue is $-\underline{2}$

$\alpha = -3$

$SI - A = \begin{bmatrix} s+3 & 2 \\ -1 & s \end{bmatrix}$

$$[SI - A]^{-1} = \frac{\begin{bmatrix} s & -2 \\ 1 & s+3 \end{bmatrix}}{(s+1)(s+2)}$$

$$X(s) = (SI - A)^{-1} X(0)$$

$$Y(s) = CX(s) = \frac{4}{s+1} = \frac{\begin{bmatrix} -1 & 1 \end{bmatrix} \begin{bmatrix} s & -2 \\ 1 & s+3 \end{bmatrix} \begin{bmatrix} x_1(0) \\ x_2(0) \end{bmatrix}}{(s+1)(s+2)}$$

Comp. the coeffs.

$$4s + 8 = x_1(0) + 5x_2(0) + (-x_1(0) + x_2(0))s$$

$$\begin{aligned} x_1(0) + 5x_2(0) &= 8 \\ -x_1(0) + x_2(0) &= 4 \end{aligned}$$

$x_2(0) = 2, x_1(0) = -2$

$X(0) = \begin{bmatrix} -2 \\ 2 \end{bmatrix}$

Similarly when $X(0) = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$, $C = \begin{bmatrix} -8/3 & -4/3 \end{bmatrix}$

$\downarrow C_2 \quad C_1$

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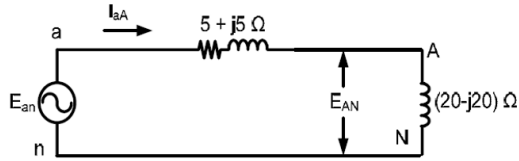
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6. Only one wattmeter is used. The question is to find the wattmeter reading. Nothing To do with total power.

$E_{AN} = 250 \angle 0^\circ$, $E_{BN} = 250 \angle -120^\circ$ and
 $E_{CN} = 250 \angle 120^\circ$
 Converting Δ into Y, $Z_Y = (60-j60)/3$
 $= (20-j20) \Omega/\text{phase}$

Single phase diagram of the circuit is



$$I_{aA} = \frac{E_{an}}{Z_{aN}} = \frac{250 \angle 0^\circ}{(5+j5) + (20-j20)} = \frac{250 \angle 0^\circ}{(5+j5) + (20-j20)} = \frac{250 \angle 0^\circ}{(25-j15)} = 7.35 + j4.41$$

$$= 8.57 \angle 31^\circ \text{ A} \quad \text{--- (1) --- 3 marks}$$

$$I_{AB} = \frac{8.57}{\sqrt{3}} \angle 31^\circ + 30^\circ = 4.95 \angle 61^\circ$$

$$I_{BC} = 4.95 \angle -59^\circ \text{ A} \quad \text{--- (2) --- 3 marks}$$

$$E_{AB} = I_{AB} \times 60 - j60 = I_{AB} \times 84.25 \angle -45^\circ$$

$$= 420 \angle 16^\circ \text{ V} \quad \text{--- (3) --- 4 marks}$$

$$W = V_{xy} \times I_{an} \cos \phi$$

$$V_{xy} = \frac{V_{an}}{\sqrt{3}} \angle 30^\circ = 433 \angle 30^\circ \text{ V}$$

$$W = 433 \times 8.57 \cos(30^\circ - 31^\circ)$$

$$= 3711 \text{ W} \quad \text{--- 5 marks}$$

7. (a) Use direct formula. See slides

$$P_{AV} = \frac{1.2 \times 0.6}{2} \cos 30^\circ = 0.31177 \text{ W}$$

$$\text{THD} = \frac{\sqrt{(0.1)^2 + (6.1)^2}}{0.6} \times 100 = \frac{6.108}{0.6} \times 100 = 1018\%$$

$$\text{True PS} = \frac{I_{s1}}{I_s} \cos \phi = \frac{1.6}{\sqrt{0.6^2 + 1.1^2}} \cos 30^\circ = 0.843$$

(b) In this question, Fig a is mentioned. The answer is not a general proof. For the graph, you have to show. In graph theory we use matrices. Branch voltages (matrix v_b) and tree branch voltages (matrix v_t) are given. How these two matrices are related.

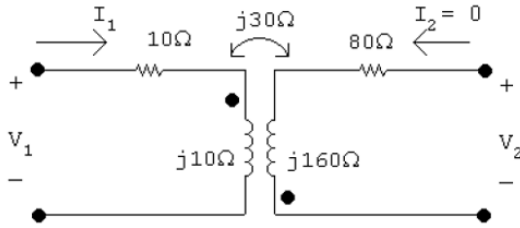
$$\text{Either } v_b = Q^T v_t \text{ or } v_b = -B^T v_t$$

$$BV = 0 \quad \times$$

$$v_b = A^T v_t \quad \times$$

$$v_b = \bar{A}^T v_n \quad \text{This is correct but not asked}$$

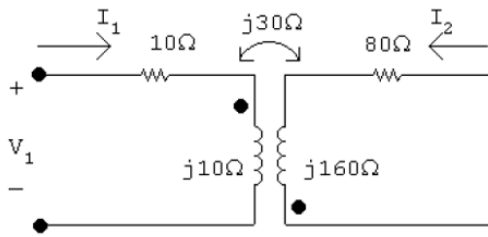
(4)

For $I_2 = 0$:

$$V_2 = -j30I_1 = -j30 \frac{V_1}{10 + j10} = \frac{-j3V_1}{1 + j1}$$

$$a_{11} = \frac{V_1}{V_2} \Big|_{I_2=0} = \frac{1 + j1}{-j3} = \frac{-1 + j1}{3}$$

$$a_{21} = \frac{I_1}{V_2} \Big|_{I_2=0} = \frac{1}{-j30} = \frac{j}{30} \text{ S}$$

For $V_2 = 0$:

$$V_1 = (10 + j10)I_1 - j30I_2$$

$$0 = -j30I_1 + (80 + j160)I_2$$

$$\Delta = \begin{vmatrix} 10 + j10 & -j30 \\ -j30 & 80 + j160 \end{vmatrix} = 100(1 + j24)$$

$$N_2 = \begin{vmatrix} 10 + j10 & V_1 \\ -j30 & 0 \end{vmatrix} = j30V_1$$

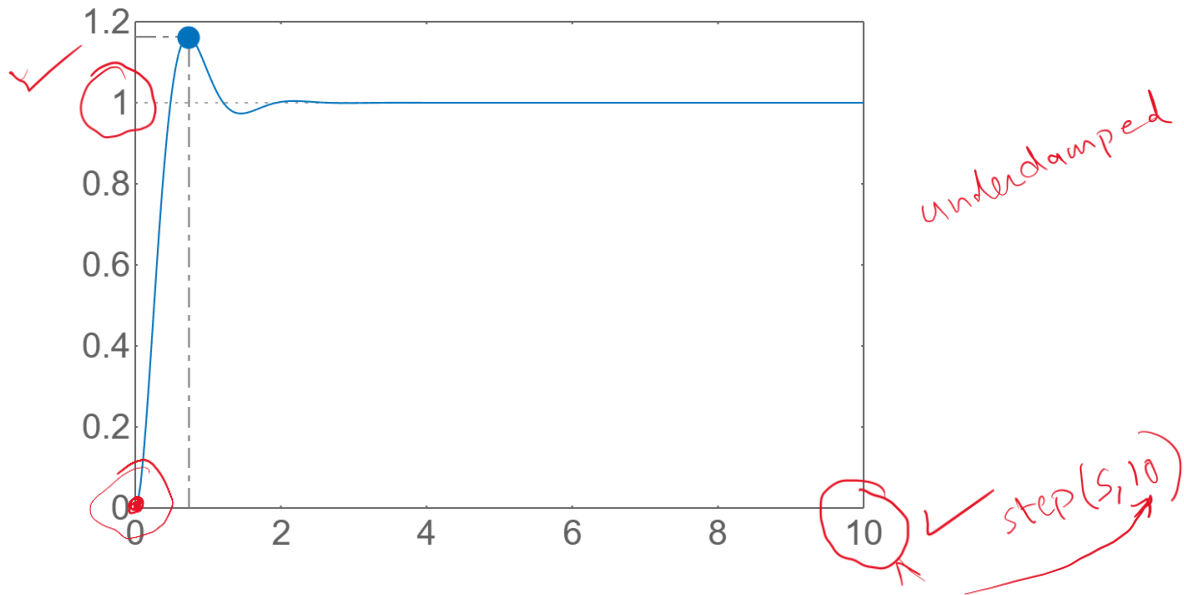
$$I_2 = \frac{N_2}{\Delta} = \frac{j30V_1}{100(1 + j24)}$$

$$a_{12} = \frac{-V_1}{I_2} \Big|_{V_2=0} = -80 + j\frac{10}{3} \Omega$$

$$j30I_1 = (80 + j160)I_2$$

$$a_{22} = -\frac{I_1}{I_2} \Big|_{V_2=0} = -\frac{8}{3}(2 - j1)$$

Question Number 4 again. ω_n was different. $\omega_n = 2, 3, 5$ and 6 rad/sec
Damping factor was same in all the sets 0.5 . Time constant was different. $1, 2/3, 0.4, 1/3$
Steady state value is 1 for all the sets.



i) The above figure only 3 marks.
Figure plotted for 10 sec, final value - 1 starting 0
underdamped.

2) time constant - 2
3) steady state - 2

4) is same as part (i). Part (i) is only figure is needed. In part (iv), you have to explain how did you get the figure without solving the equation. 4 marks